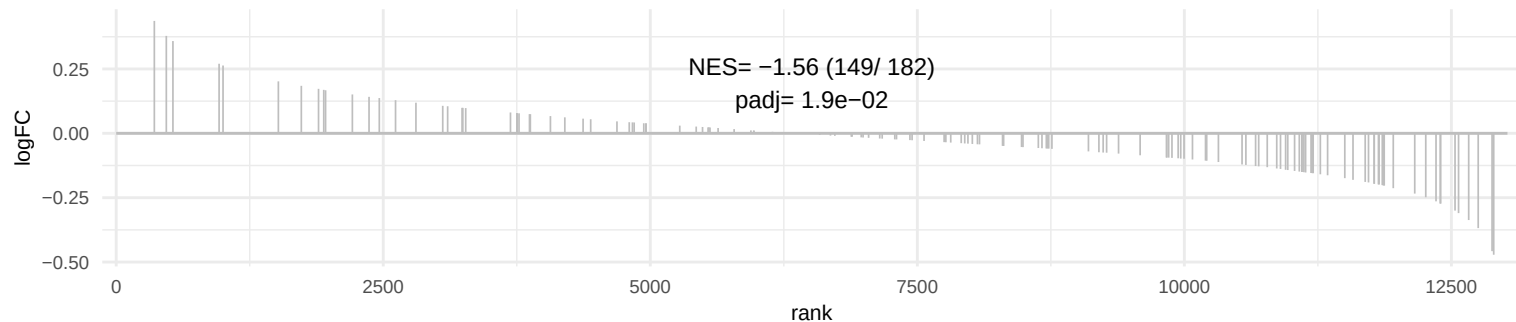


Neurotransmitter Receptors And Postsynaptic Signal Transmission R-HSA-112314

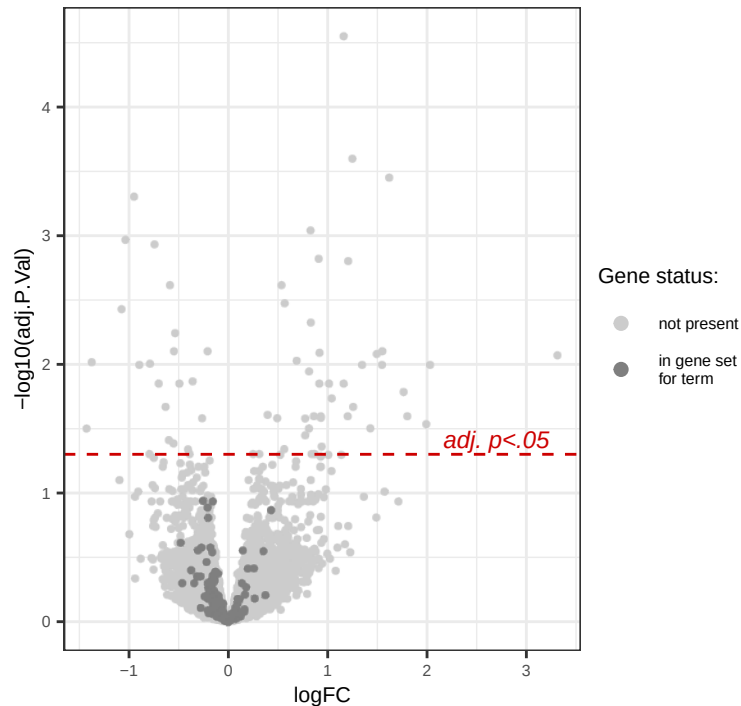


Since fgsea results are calculated based on the ranked $\log FC$ values, many fgsea results with $padj < .05$ do not contain any L-R DEGs. For our analysis, we have filtered out such results.

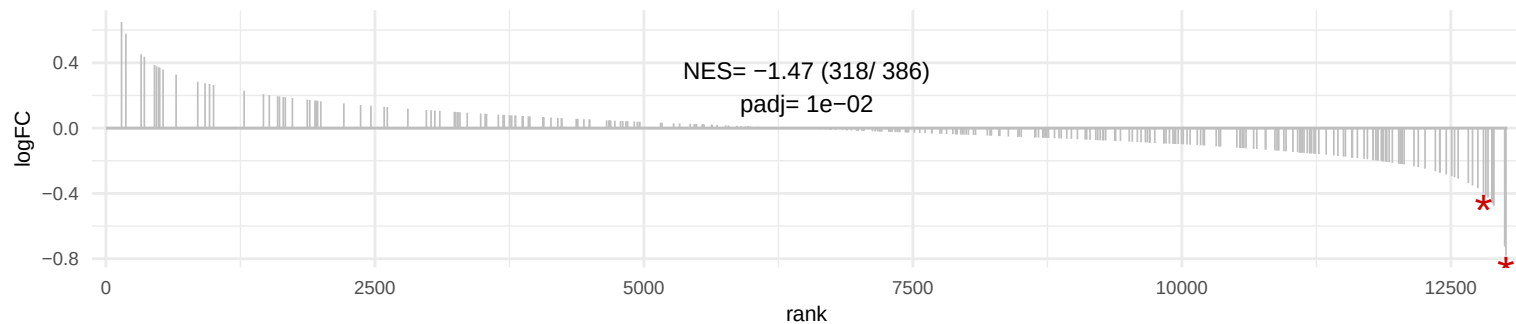
fgsea manhattan plot (top): Lowly ranked values (x-axis) have the highest, most positive $\log FC$ (y-axis).

Volcano plot (right): Opposite from above, the $\log FC$ (x-axis) shows negative values on left and positive values on right.

Volcano showing genes contributing to GSEA result



Neuronal System R-HSA-112316

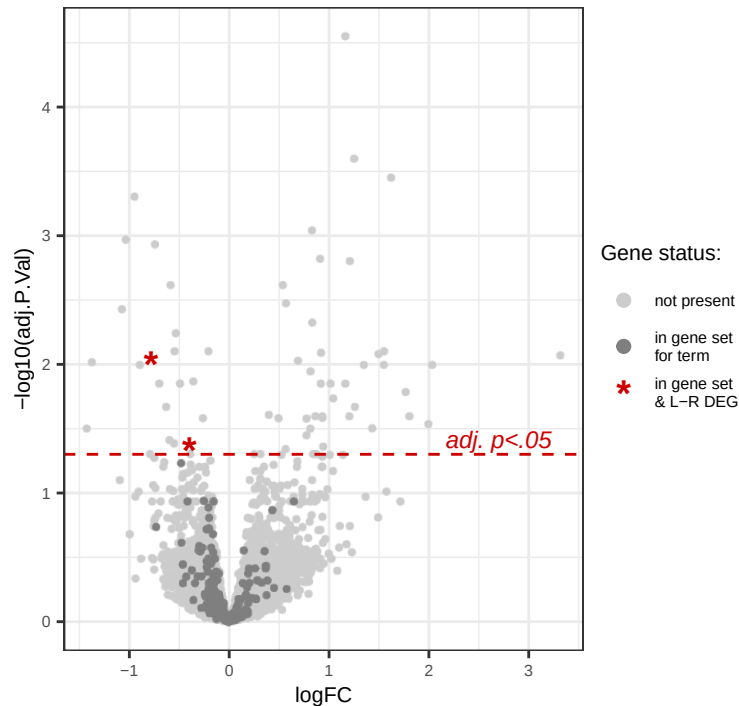


Significant ($\text{padj} < .05$) fgsea results were filtered to those that contained a L-R DEG in the leading edge genes. These L-R DEGs did not have to be paired (sig. in both annotations), allowing for unique PRECAST cluster or Seurat cell type label results to emerge (if present).

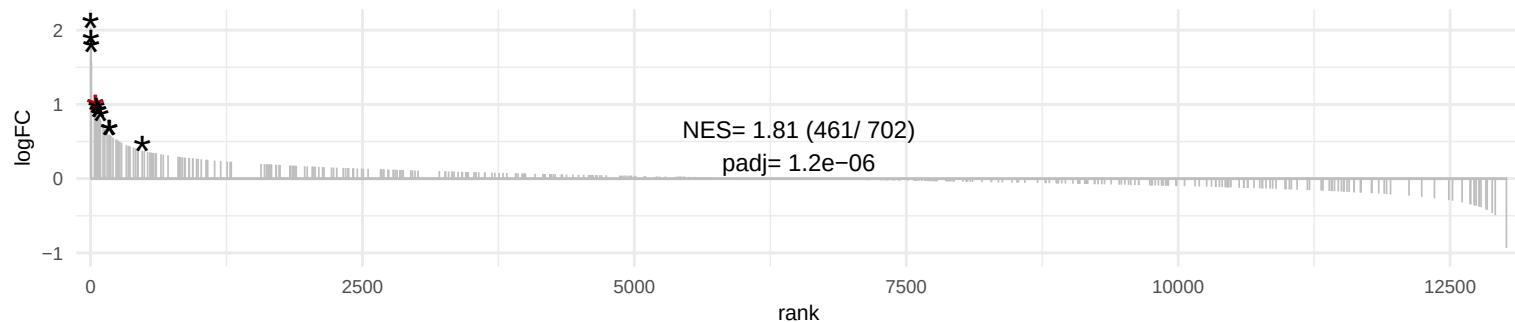
fgsea manhattan plot (top): The rank position of L-R DEGs is indicated with a red asterisk.

Volcano plot (right): The L-R DEGs are indicated with a red asterisk.

Volcano showing genes contributing to GSEA result



Cytokine Signaling In Immune System R-HSA-1280215

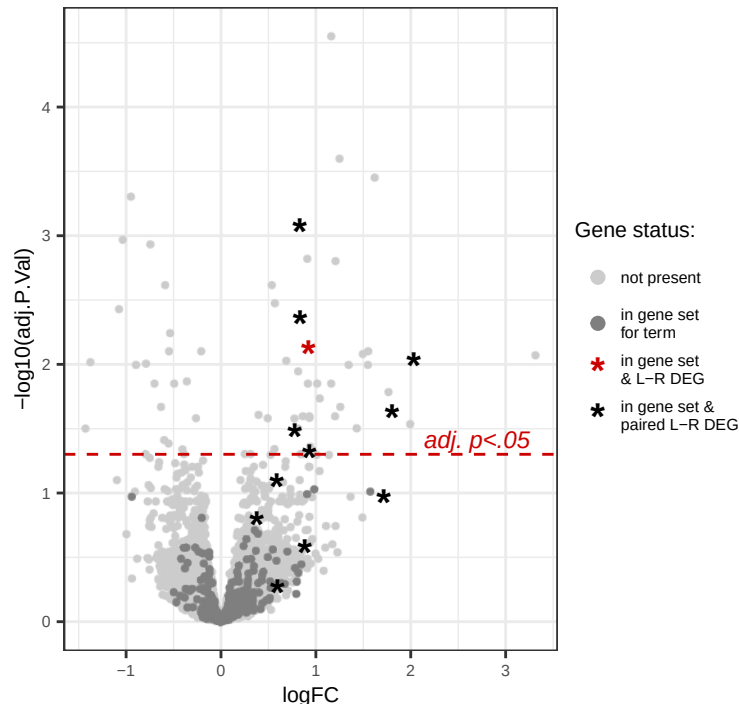


Significant ($\text{padj} < .05$) fgsea results containing a L-R DEG were further stratified by those results that also contain a paired L-R DEG in the leading edge genes. This will help distinguish the smaller number of sig. results that only contained layer-specific L-R DEGs that were not reproducible in the other annotation strategy.

fgsea manhattan plot (top): The rank position of paired L-R DEGs is indicated with a black asterisk. Any additional layer-specific L-R DEGs are also indicated with a red asterisk.

Volcano plot (right): In most cases, filtering by paired L-R DEGs is more restrictive (generates fewer results). However, for the individual PRECAST clusters/ Seurat labels that comprise L2_L3_L4 like this example, our approach of looking at the union of L-R DEGs within a single annotation means that some paired L-R DEGs (black asterisk) are not statistically significant in the group for which the fgsea was performed (see black asterisks below the dashed red line).

Volcano showing genes contributing to GSEA result



NTC.MDD F

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	42	68	21	56	18
smoothed	L2_L3_L4	L3.4	47	55	20	56	20
seurat	L2_L3_L4	L2.3	48	80	28	56	23
seurat	L2_L3_L4	L4	47	61	18	56	17
smoothed	L5	L5	49	51	20	44	20
seurat	L5	L5	53	68	21	44	20
smoothed	L6	L6	54	60	26	46	24
seurat	L6	L6	59	68	27	46	27

NTC.MDD M

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	41	0	0	0	0
smoothed	L2_L3_L4	L3.4	87	0	0	0	0
seurat	L2_L3_L4	L2.3	53	0	0	0	0
seurat	L2_L3_L4	L4	38	0	0	0	0
smoothed	L5	L5	120	0	0	0	0
seurat	L5	L5	140	0	0	0	0
smoothed	L6	L6	89	0	0	0	0
seurat	L6	L6	77	0	0	0	0

NTC.BPD F

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	38	23	13	21	13
smoothed	L2_L3_L4	L3.4	33	12	8	21	8
seurat	L2_L3_L4	L2.3	37	32	10	21	10
seurat	L2_L3_L4	L4	30	16	5	21	5
smoothed	L5	L5	41	32	11	26	11
seurat	L5	L5	33	41	9	26	9
smoothed	L6	L6	40	77	34	42	10
seurat	L6	L6	39	56	10	42	10

NTC.BPD M

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	33	51	14	34	9
smoothed	L2_L3_L4	L3.4	12	38	6	34	5
seurat	L2_L3_L4	L2.3	22	44	13	34	8
seurat	L2_L3_L4	L4	5	9	0	34	0
smoothed	L5	L5	6	28	2	17	2
seurat	L5	L5	9	23	4	17	4
smoothed	L6	L6	4	15	2	8	2
seurat	L6	L6	9	13	5	8	5

MDD.BPD F

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	30	11	19	9	15
smoothed	L2_L3_L4	L3.4	34	3	0	9	0
seurat	L2_L3_L4	L2.3	29	13	13	9	13
seurat	L2_L3_L4	L4	23	3	0	9	0
smoothed	L5	L5	22	0	0	0	0
seurat	L5	L5	22	4	0	0	0
smoothed	L6	L6	18	11	1	8	0
seurat	L6	L6	19	12	2	8	2

MDD.BPD M

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	55	28	7	38	6
smoothed	L2_L3_L4	L3.4	86	44	23	38	20
seurat	L2_L3_L4	L2.3	62	46	15	38	9
seurat	L2_L3_L4	L4	45	62	17	38	8
smoothed	L5	L5	71	18	10	14	4
seurat	L5	L5	93	33	14	14	6
smoothed	L6	L6	64	2	0	2	0
seurat	L6	L6	63	13	7	2	0